

■ COMPLETE NOTES — EXPANDED EDITION

HTML

Basic Structure + Advanced Topics

CodeWithHarry — Sigma Web Development Course

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HTML Boilerplate (Basic Structure)

Har HTML file ka skeleton — pehle yahi likhte hain

KYA HAI? HTML file ka basic template jo browser ko batata hai ki yeh ek HTML document hai. Bina iske browser sahi se page nahi samajhta.

```
<!-- Ye line browser ko batati hai: 'Ye HTML5 document hai' -->
<!DOCTYPE html>
<html lang='en'>
  <head>
    <meta charset='UTF-8'>
    <title>Meri Pehli Website</title>
  </head>
  <body>
    <!-- Yahan aata hai jo user screen par dekhta hai -->
  </body>
</html>
```

<!DOCTYPE html>	HTML version declare karta hai (HTML5)
<html>	Root element — saari cheez iske andar
<head>	Metadata — title, charset, CSS links (user nahi dekhta)
<title>	Browser tab mein naam dikhata hai
<body>	Yahan ka content screen par dikhta hai



Headings — <h1> se <h6>

6 levels ke headings hote hain — h1 sabse bada, h6 sabse chota

KYA HAI? Headings text ko important aur bada banane ke kaam aate hain. Newspaper ki tarah — ek main headline, phir sub-headings.

■ **SEO:** h1 ek page mein sirf ek baar hona chahiye — Google ise sabse important maanta hai.

```
<h1>Sabse Bada Heading (Main Title)</h1>
<h2>Dusra Heading (Section Title)</h2>
<h3>Teesra Heading (Sub-section)</h3>
<h4>Chautha Heading</h4>
<h5>Paanchwa Heading</h5>
<h6>Chata Heading (Sabse Chota)</h6>
```

■ **Tip:** Headings sirf size ke liye mat use karo — structure ke liye use karo. Size baad mein CSS se change hongii.



Paragraph — <p>

Normal text content likhne ke liye

KYA HAI? Paragraph tag se normal text likha jaata hai. Browser automatically upar-neeche thodi space (margin) de deta hai.

```
<p>Yeh mera pehla paragraph hai. Main HTML seekh raha hoon.</p>
<p>Yeh dusra paragraph hai. Dono ke beech automatically gap aa jaata hai.</p>

<!-- Special tags jo paragraph ke saath use hote hain -->
<p>Yeh <b>bold</b> hai aur yeh <i>italic</i> hai.</p>
<p>Line break ke liye: Pehli line<br>Dusri line</p>
<hr> <!-- Horizontal line -->
```

	Bold text
<i>	Italic text
<u>	Underline text
 	Line break (self-closing)
<hr>	Horizontal divider line



Links (Anchor Tag) — <a>

Ek page se dusre page ya website par jaane ke liye

KYA HAI? Anchor tag se hyperlink banate hain. href attribute mein URL dete hain jahan click karne par jana ho.

```
<!-- Kisi website par jaana -->
<a href='https://www.google.com'>Google par jao</a>

<!-- Naye tab mein kholna -->
<a href='https://www.youtube.com' target='_blank'>YouTube (Naye tab mein)</a>

<!-- Apne page ke kisi section par jaana -->
<a href='#contact'>Contact section par jao</a>

<!-- Email link -->
<a href='mailto:harry@example.com'>Email Karo</a>
```

href	Kahan jaana hai — URL ya #section
target='_blank'	Link naye tab mein khulega
target='_self'	Same tab mein (default)
mailto:	Email app open karta hai



Images —

Webpage par image dikhane ke liye (self-closing tag)

KYA HAI? img tag se webpage par image show karte hain. Yeh self-closing tag hai — iska closing tag nahi hota.

■ **ZARURI:** alt attribute hamesha likho — image na load ho tab text dikhta hai, aur visually impaired users ke liye bhi zaroori hai.

```
<!-- Local image (apne folder se) -->
<img src='meri-photo.jpg' alt='Meri Photo'>

<!-- Internet se image -->
<img src='https://example.com/image.jpg'
      alt='Example Image' width='400' height='300'>

<!-- Image with link (clickable image) -->
<a href='https://google.com'>
  <img src='logo.png' alt='Google Logo'>
</a>
```

src	Image ka path ya URL
alt	Image na aaye toh alternate text
width/height	Size pixels mein (CSS se better control)



Lists — , , <dl>

3 tarah ki lists hoti hain HTML mein

UL — Unordered

OL — Ordered

DL — Description

```
<!-- Unordered List (Bullet) -->
<ul>
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ul>

<!-- Ordered List (Numbered) -->
<ol>
  <li>HTML seekho</li>
  <li>CSS seekho</li>
  <li>JavaScript seekho</li>
</ol>

<!-- Description List -->
<dl>
  <dt>HTML</dt>
  <dd>HyperText Markup Language</dd>
  <dt>CSS</dt>
  <dd>Cascading Style Sheets</dd>
</dl>
```

■ **Tip:** List ke andar list bhi bana sakte ho — ise Nested List kehte hain! ke andar ek aur ya daal do.



Tables — <table>

Row aur column mein data dikhane ke liye

KYA HAI? Table mein rows hoti hain (<tr>), aur rows mein cells. Header cell = <th>, Normal cell = <td>

```
<table border='1'>
  <!-- Table Head -->
  <thead>
    <tr><th>Naam</th><th>Subject</th><th>Marks</th></tr>
  </thead>
  <!-- Table Body -->
  <tbody>
    <tr><td>Rohan</td><td>Maths</td><td>95</td></tr>
    <tr><td>Priya</td><td>Science</td><td>88</td></tr>
  </tbody>
  <!-- Table Footer -->
  <tfoot>
    <tr><td colspan='2'>Total</td><td>183</td></tr>
  </tfoot>
</table>
```

<table>	Table ka wrapper
<tr>	Table Row — ek puri line
<th>	Header cell (bold + centered)
<td>	Normal data cell
colspan	Ek cell multiple columns span kare
rowspan	Ek cell multiple rows span kare

■ **Tip:** Aaj kal tables sirf tabular data ke liye use hoti hain, layout ke liye nahi. Layout CSS se karte hain!



SEO & Core Web Vitals in HTML

Search Engine Optimization aur Performance Metrics

KYA HAI? SEO (Search Engine Optimization) matlab apne page ko Google mein upar laana. HTML mein kuch special tags hote hain jo search engines ko page samajhne mein help karte hain.

■ Essential Meta Tags

```
<head>
  <!-- Page title – Google SERP mein dikhta hai -->
  <title>Sigma Web Dev | HTML Notes</title>

  <!-- Page ka short description (150-160 chars) -->
  <meta name='description'
        content='Complete HTML notes for beginners – CodeWithHarry.'>

  <!-- Keywords (modern Google zyada nahi dekhta, phir bhi likho) -->
  <meta name='keywords' content='HTML, CSS, web development'>

  <!-- Mobile responsive banana -->
  <meta name='viewport' content='width=device-width, initial-scale=1.0'>

  <!-- Canonical URL – duplicate content problem solve karta hai -->
  <link rel='canonical' href='https://yoursite.com/html-notes'>

  <!-- Robots ko instructions -->
  <meta name='robots' content='index, follow'>
</head>
```

■ Open Graph Tags (Social Media Preview)

```
<!-- Facebook / WhatsApp / LinkedIn preview ke liye -->
<meta property='og:title' content='HTML Complete Notes'>
<meta property='og:description' content='Sigma Web Dev – CodeWithHarry'>
<meta property='og:image' content='https://yoursite.com/preview.jpg'>
<meta property='og:url' content='https://yoursite.com/html-notes'>
<meta property='og:type' content='website'>

<!-- Twitter Card -->
<meta name='twitter:card' content='summary_large_image'>
<meta name='twitter:title' content='HTML Complete Notes'>
<meta name='twitter:image' content='https://yoursite.com/preview.jpg'>
```

■ Semantic HTML — SEO ka backbone

KYA HAI? Semantic tags search engines ko page ka structure clearly batate hain. <div> ki jagah semantic tags use karo — Google inhe zyada pasand karta hai.

```
<header>    <!-- Page ka top section, logo, nav -->
  <nav>      <!-- Navigation links -->
    <a href='/'>Home</a>
    <a href='/about'>About</a>
  </nav>
</header>

<main>      <!-- Page ka main content -->
  <article>  <!-- Self-contained content (blog post etc.) -->
    <h1>Main Title – Sirf ek baar per page!</h1>
    <section> <!-- Ek related content group -->
      <h2>Sub Section</h2>
      <p>Content here...</p>
    </section>
  </article>
  <aside>    <!-- Sidebar / related content -->
    <p>Related Links</p>
  </aside>
</main>

<footer>    <!-- Page ka bottom section -->
  <p>Copyright 2024</p>
</footer>
```

<header>	Page/section ka top area
<nav>	Navigation links ka group
<main>	Page ka primary content (sirf ek baar)
<article>	Independent, self-contained content
<section>	Thematically grouped content
<aside>	Sidebar ya supplementary content
<footer>	Page/section ka bottom area

■ Core Web Vitals — Kya Hote Hain?

KYA HAI? Core Web Vitals Google ke 3 main performance metrics hain jo directly ranking affect karte hain. Inhe 2021 se Google officially ranking factor maanta hai.

Metric	Full Name	Kya Measure Karta Hai	Good Score
LCP	Largest Contentful Paint	Sabse bada element kitni jaldi load hota hai	≤ 2.5 sec



Forms & Input Tags in HTML

User se data collect karne ke liye — login, signup, contact forms

KYA HAI? <form> tag ek container hai jisme saare input elements aate hain. Form submit hone par data server par jaata hai ya JavaScript handle karta hai.

Basic Form Structure

```
<form action='/submit' method='POST'>

  <!-- Text input -->
  <label for='name'>Aapka Naam:</label>
  <input type='text' id='name' name='name'
        placeholder='Naam likhein' required>

  <!-- Email input -->
  <label for='email'>Email:</label>
  <input type='email' id='email' name='email' required>

  <!-- Password -->
  <input type='password' id='pass' name='password'
        minlength='8' placeholder='Min 8 characters'>

  <!-- Submit button -->
  <button type='submit'>Submit Karo</button>

</form>
```

All Input Types — Comprehensive List

Type	Code Example	Use Case
text	type="text"	Normal text input
email	type="email"	Email — format auto-validate
password	type="password"	Password — text hidden hota hai
number	type="number" min="1" max="100"	Numeric input with range
tel	type="tel"	Phone number
url	type="url"	Website URL
date	type="date"	Date picker
time	type="time"	Time picker
checkbox	type="checkbox"	Multiple options select

radio	<code>type="radio" name="grp"</code>	Ek option select (same name group)
range	<code>type="range" min="0" max="100"</code>	Slider input
color	<code>type="color"</code>	Color picker
file	<code>type="file" accept=".pdf,.jpg"</code>	File upload
hidden	<code>type="hidden" value="xyz"</code>	User ko nahi dikhta, form ke saath submit hota hai
submit	<code>type="submit"</code>	Form submit karta hai
reset	<code>type="reset"</code>	Form clear karta hai

■ Other Form Elements

```

<!-- Textarea: Multi-line text input -->
<textarea name='message' rows='5' cols='40'
        placeholder='Apna message likhein...'></textarea>

<!-- Select Dropdown -->
<select name='city'>
  <option value=''>-- Sheher chunein --</option>
  <option value='delhi'>Delhi</option>
  <option value='mumbai' selected>Mumbai</option>
  <option value='pune'>Pune</option>
</select>

<!-- Datalist: Input + Suggestions combo -->
<input list='frameworks' name='framework' placeholder='Framework...'>
<datalist id='frameworks'>
  <option value='React'>
  <option value='Vue'>
  <option value='Angular'>
</datalist>

<!-- Fieldset: Related inputs group karo -->
<fieldset>
  <legend>Personal Info</legend>
  <input type='text' name='fname' placeholder='First Name'>
  <input type='text' name='lname' placeholder='Last Name'>
</fieldset>

```

■ Important Form Attributes

action	Form data kahan bheja jaaye — URL ya server endpoint
method	GET (URL mein data) ya POST (hidden, secure — use this)
required	Field khali nahi chhod sakte — browser validate karta hai
placeholder	Input mein hint text — submit hone par clear ho jaata hai

value	Default value jo input mein pehle se hogi
disabled	Input clickable nahi hogi — form ke saath submit bhi nahi hogi
readonly	Input dikhegi but edit nahi kar sakte
min / max	Number aur date inputs ke liye range set karo
maxlength	Maximum characters allowed
autocomplete	Browser ka auto-fill on/off karo
autofocus	Page load hote hi yeh input focus pe hogi
multiple	File/email input mein multiple values allow karo
pattern	Regex pattern se input validate karo
name	Form submit hone par is naam se data jaata hai — ZAROORI hai
id + for	Label aur input ko link karna — accessibility ke liye

■ **Tip:** Hamesha `<label>` use karo aur uska `'for'` attribute input ke `'id'` se match karo. Isse screen readers aur click area dono improve hote hain!



Inline & Block Elements in HTML

HTML ke do fundamental display types — sabhi elements inka hissa hain

KYA HAI? HTML ka har element ya toh Block-level hota hai ya Inline. Yeh decide karta hai ki element kitni jagah leta hai aur kahan display hota hai.

Property	Block Elements	Inline Elements
Width	Full available width leta hai	Sirf content jitni width
Height	Content + padding ke hisaab se	Content ke hisaab se
New Line	Haan — naya line shuru karta hai	Nahi — same line mein rehta hai
Width/Height CSS	CSS se set ho sakti hai	Width/height CSS se set NAHI hoti
Margin/Padding	Chaaron taraf lagta hai	Sirf left/right (top/bottom limited)
Nesting	Block + inline dono contain kar sakta hai	Sirf inline elements contain kar sakta hai
Default Display	CSS: display: block	CSS: display: inline

■ BLOCK Elements — Comprehensive List

```
<!-- Layout & Structure -->
<div>    Generic block container</div>
<header> Page/section ka header</header>
<footer> Page/section ka footer</footer>
<main>    Primary content area</main>
<nav>    Navigation links</nav>
<section> Thematically grouped content</section>
<article> Self-contained content</article>
<aside>    Sidebar / supplementary content</aside>

<!-- Text & Headings -->
<h1> to <h6>    Headings (6 levels)
<p>            Paragraph</p>
<blockquote>    Long quoted text</blockquote>
<pre>            Preformatted text (whitespace preserved)</pre>
<address>        Contact information</address>
<figcaption>    Caption for a figure</figcaption>

<!-- Lists -->
<ul>    Unordered list</ul>
<ol>    Ordered list</ol>
<li>    List item</li>
<dl>    Description list</dl>
<dt>    Term in description list</dt>
<dd>    Definition/description</dd>

<!-- Media & Embed -->
<figure>    Image + caption wrapper</figure>
<video>    Video player</video>
<audio>    Audio player</audio>
<canvas>    Drawing surface (JS ke saath)</canvas>
<table>    Table</table>
<hr>        Horizontal rule (self-closing)

<!-- Form -->
<form>        Form container</form>
<fieldset>    Related inputs group</fieldset>
```

■ INLINE Elements — Comprehensive List

```
<!-- Text Formatting -->
<span>    Generic inline container – CSS ke liye</span>
<a>      Hyperlink</a>
<strong> Important text (bold + semantic)</strong>
<b>      Bold (sirf visual, no meaning)</b>
<em>      Emphasized text (italic + semantic)</em>
<i>      Italic (sirf visual)</i>
<u>      Underline</u>
<s>      Strikethrough</s>
<mark>   Highlighted text (yellow bg default)</mark>
<small>   Smaller font size text</small>
<big>    Bigger font size text</big>

<!-- Subscript / Superscript -->
<sub>    Subscript: H<sub>2</sub>O</sub>
<sup>    Superscript: x<sup>2</sup></sup>

<!-- Code & Technical -->
<code>   Inline code snippet</code>
<kbd>    Keyboard input – Ctrl+C like</kbd>
<samp>    Sample output</samp>
<var>    Variable name</var>
<abbr>   Abbreviation: <abbr title='HTML'>HTML</abbr>
<cite>   Citation / reference</cite>
<q>      Short inline quote</q>

<!-- Self-Closing Inline -->
<img src='...' alt='...'>    Image
<br>                          Line break
<input type='text'>          Form input
<label>  Form label</label>
<button> Clickable button</button>
<select> Dropdown</select>
<textarea>Multi-line input</textarea>
```

■ display: inline-block — Best of Both Worlds

KYA HAI? inline-block ek CSS display value hai jo inline element ki tarah same line mein rehta hai, lekin block element ki tarah width/height bhi set ho sakti hai.


```

/* CSS mein: */
.badge {
  display: inline-block;
  width: 100px;      /* inline pe kaam nahi karta — yahan karta hai! */
  height: 30px;
  padding: 5px 10px;
  background: #0f3460;
}

<!-- HTML: -->
<span class='badge'>HTML</span>
<span class='badge'>CSS</span>
<span class='badge'>JS</span>
<!-- Teeno same line mein honge, lekin custom size ke saath -->

```

display: block	Element puri width leta hai, naya line shuru hota hai
display: inline	Same line mein, width/height set nahi hoti
display: inline-block	Same line mein, lekin width/height bhi set hoti hai
display: none	Element page se completely remove ho jaata hai
display: flex	Flexbox layout — modern alignment ke liye
display: grid	Grid layout — 2D layouts ke liye

■ Common Mistakes & Rules

■ **Block element ko inline ke andar NAHI rakhte** — jaise `<div>` ko `<span>` ke andar ya `<p>` ko `<a>` ke andar.

■ **Tip:** Agar sirf styling ke liye container chahiye — `<div>` (block) ya `<span>` (inline) use karo, semantic meaning nahi hota inka.

```

<!-- GALAT: Block inside inline -->
<span><div>Yeh wrong hai</div></span>
<a><p>Yeh bhi wrong hai</p></a>

<!-- SAHI: Inline inside block -->
<div><span>Yeh correct hai</span></div>
<p>Yeh <strong>bhi</strong> correct hai.</p>

<!-- SAHI: Block ke andar block -->
<div>
  <h2>Heading</h2>
  <p>Paragraph</p>
</div>

```

■ **Tip:** HTML5 mein `<a>` tag ke andar block elements allowed hain — jaise ek pura card clickable banana. Lekin avoid karo jab tak zaroorat na ho.



ID and Classes in HTML

Elements ko target karne ke do powerful tarike — CSS aur JS ke liye zaroori

KYA HAI? ID aur Class HTML attributes hain jo elements ko uniquely identify karne ya group karne ke liye use hote hain. CSS styling aur JavaScript dono inhi pe depend karte hain.

■ ID — Unique Identifier

■ ID ek page pe sirf BAAR use hoti hai — ek ID = ek element. CSS mein # se target karte hain.

```
<!-- HTML mein ID dena -->
<div id="main-header">Yeh page ka header hai</div>
<p id="intro-text">Yeh intro paragraph hai</p>
<button id="submit-btn">Submit</button>

/* CSS mein ID target karna — # use karo */
#main-header {
  background: #0f3460;
  color: white;
  padding: 20px;
}

// JavaScript mein ID se element pakadna
const btn = document.getElementById("submit-btn");
btn.addEventListener("click", function() {
  alert("Button clicked!");
});
```

■ Class — Reusable Group

KYA HAI? Class ek page pe multiple elements pe use ho sakti hai. Ek element ke multiple classes bhi ho sakti hain. CSS mein . se target karte hain.

```

<!-- Ek class multiple elements pe -->
<p class="highlight">Yeh highlighted hai</p>
<p class="highlight">Yeh bhi highlighted hai</p>
<h2 class="highlight">Heading bhi highlighted</h2>

<!-- Ek element pe multiple classes -->
<div class="card active featured">
  Yeh div teen classes use kar raha hai!
</div>

/* CSS mein class target karna — . use karo */
.highlight {
  background: #ffd166;
  color: #1a1a2e;
  padding: 5px 10px;
  border-radius: 4px;
}

.card { border: 1px solid #0f3460; padding: 20px; }
.active { border-color: #06d6a0; }
.featured { background: #16213e; }

// JavaScript mein class se elements pakadna
const items = document.querySelectorAll(".highlight");
items.forEach(el => el.style.color = 'red');

```

■ ID vs Class — Kab Kya Use Karein?

Property	ID (#)	Class (.)
Uniqueness	Sirf ek element per page	Multiple elements pe use ho sakti hai
CSS Selector	#myId { }	.myClass { }
JS Selector	getElementById('myId')	querySelectorAll('.myClasses')
Specificity	Zyada specific — class ko override karta hai	Kam specific — easily override hoti hai
Reusability	Nahi — ek baar hi use karo	Haan — jitni baar chahein
Anchor Link	Haan! href='#section-id' se jump karo	Nahi hota
Best Use	Unique sections: #navbar, #footer, #hero	Reusable styles: .btn, .card, .active

■ CSS Specificity — Priority Order

KYA HAI? Jab multiple CSS rules ek element pe apply hoti hain, browser decide karta hai kaunsi rule jeetegi — ise Specificity kehte hain.

```
/* Specificity order (low to high): */

/* 1. Element selector – sabse kam specific */
p { color: white; }           /* specificity: 0,0,1 */

/* 2. Class selector */
.highlight { color: yellow; } /* specificity: 0,1,0 */

/* 3. ID selector – sabse specific */
#special { color: red; }      /* specificity: 1,0,0 */

/* 4. Inline style – ID se bhi zyada */
<p style="color: green;">     /* specificity: 1,0,0,0 */

/* 5. !important – sabka baap, use wisely! */
p { color: pink !important; }
```

■ **Tip:** Best practice: Classes use karo for styling, IDs use karo sirf JavaScript targeting aur page anchor links ke liye. !important se bachne ki koshish karo — maintenance nightmare ban jaata hai!

■ Naming Conventions — Best Practices

```
<!-- GALAT naming -->
<div id="1section">           <!-- Number se shuru nahi hona chahiye -->
<div class="My Class">        <!-- Spaces allowed nahi hain -->
<div id="a">                   <!-- Too short, meaningless -->

<!-- SAHI naming -->
<div id="main-header">         <!-- kebab-case: words with hyphens -->
<div class="nav-link active">
<section id="about-section">

<!-- BEM Naming Convention (popular method) -->
<!-- Block__Element--Modifier -->
<div class="card">
  <h2 class="card__title">Title</h2>
  <p class="card__text">Content</p>
  <button class="card_btn card_btn--primary">Click</button>
</div>
```

kebab-case	Words ko hyphen se jodo — main-header, nav-link (recommended)
camelCase	JavaScript mein common — mainHeader, navLink
BEM	Block__Element--Modifier — large projects ke liye best
No spaces	Class/ID name mein spaces kabhi mat dalo
Meaningful	Name se purpose clear hona chahiye — #btn nahi, #submit-btn



Video, Audio & Media in HTML

Webpage pe multimedia embed karna — natively, bina kisi plugin ke

KYA HAI? HTML5 ne native video aur audio support diya — ab Flash ya plugins ki zaroorat nahi. Sirf HTML tags se hi video/audio embed kar sakte ho.

■ Video Tag — <video>

```
<!-- Basic video -->
<video width='640' height='360' controls>
  <source src='myvideo.mp4' type='video/mp4'>
  <source src='myvideo.webm' type='video/webm'>
  Aapka browser video support nahi karta.
</video>

<!-- All attributes ke saath -->
<video
  width='100%'
  height='auto'
  controls
  autoplay
  muted
  loop
  poster='thumbnail.jpg'
  preload='metadata'>
  <source src='video.mp4' type='video/mp4'>
</video>
```

controls	Play/pause/volume controls dikhata hai
autoplay	Page load hone par video automatically play ho
muted	Video muted start ho — autoplay ke liye zaroori hai browsers mein
loop	Video end hone par dobara start ho jaaye
poster	Video load hone se pehle thumbnail image dikhao
preload	none / metadata / auto — kitna preload karna hai
width / height	Video player ka size — % ya px mein

■ Audio Tag — <audio>

```

<!-- Basic audio -->
<audio controls>
  <source src='mysong.mp3' type='audio/mpeg'>
  <source src='mysong.ogg' type='audio/ogg'>
  Aapka browser audio support nahi karta.
</audio>

<!-- All attributes ke saath -->
<audio
  controls
  autoplay
  muted
  loop
  preload='auto'>
  <source src='podcast.mp3' type='audio/mpeg'>
</audio>

```

controls	Audio player controls dikhao
autoplay	Page load hote hi play shuru
loop	Audio khatam hone par phir se chalao
muted	Muted start karo
preload	auto = pura load, metadata = sirf info, none = kuch nahi

■ Supported Media Formats

Format	Type String	Support	Best For
MP4 (H.264)	video/mp4	Chrome, Firefox, Safari, Edge	<i>Web video — most compatible</i>
WebM	video/webm	Chrome, Firefox, Edge	<i>Open format, smaller size</i>
OGG Video	video/ogg	Chrome, Firefox	<i>Open source projects</i>
MP3	audio/mpeg	Sab browsers	<i>Audio — most compatible</i>
WAV	audio/wav	Chrome, Firefox, Safari	<i>High quality, uncompressed</i>
OGG Audio	audio/ogg	Chrome, Firefox	<i>Open format audio</i>
AAC	audio/aac	Chrome, Safari, Edge	<i>iOS/Safari ke liye better</i>

■ **Tip:** Hamesha do formats provide karo — MP4 + WebM (video) ya MP3 + OGG (audio). Browser jo pehla supported format milega, woh use karega!

■ YouTube & iFrame Embed

KYA HAI? YouTube ya koi bhi external content embed karne ke liye `<iframe>` tag use hota hai. YouTube videos direct host karne ki jagah embed karna better hai — hosting free!

```
<!-- YouTube video embed -->
<iframe
  width='560'
  height='315'
  src='https://www.youtube.com/embed/VIDEO_ID'
  title='YouTube video player'
  frameborder='0'
  allow='accelerometer; autoplay; clipboard-write;
        encrypted-media; gyroscope; picture-in-picture'
  allowfullscreen>
</iframe>

<!-- Google Maps embed -->
<iframe
  src='https://maps.google.com/maps?q=Delhi&output=embed'
  width='600'
  height='450'
  style='border:0;'
  loading='lazy'
  allowfullscreen>
</iframe>

<!-- Spotify embed -->
<iframe
  src='https://open.spotify.com/embed/track/TRACK_ID'
  width='300'
  height='380'
  frameborder='0'
  allow='encrypted-media'>
</iframe>
```

src	Embed karne wali URL — YouTube/Maps/Spotify etc.
width / height	iFrame ka size pixels ya % mein
fancybox	Border dikhaao ya nahi — 0 means no border
allowfullscreen	Fullscreen button enable karo
loading='lazy'	iFrame tab load ho jab user us tak scroll kare
allow	Kaunsi features allow hain — autoplay, camera etc.
title	Accessibility ke liye — screen readers use karte hain

■ Figure & Figcaption — Semantic Media Wrapper

KYA HAI? `<figure>` tag media (image/video/code) ko wrap karta hai aur `<figcaption>` uska caption provide karta hai. Yeh semantic hai — SEO aur accessibility dono ke liye better hai.

```
<!-- Image with caption -->
<figure>
  <img src='sunset.jpg' alt='Beautiful sunset over mountains'
        width='800' height='500'>
  <figcaption>Fig 1: Sunset over the Himalayas – 2024</figcaption>
</figure>

<!-- Video with caption -->
<figure>
  <video controls width='640'>
    <source src='demo.mp4' type='video/mp4'>
  </video>
  <figcaption>Demo: HTML5 Video Tag in action</figcaption>
</figure>

<!-- Code example with caption -->
<figure>
  <pre><code>
    const x = 10;
    console.log(x);
  </code></pre>
  <figcaption>Example 1: JavaScript variable declaration</figcaption>
</figure>
```

■ **Tip:** *Figure + figcaption use karo jab media aur uska description saath mein ho. Search engines figcaption ko image description ke roop mein bhi padhte hain!*

■ Picture Tag — Responsive Images

KYA HAI? <picture> tag se alag screen sizes ke liye alag images provide kar sakte ho. Mobile pe choti image, desktop pe badi — bandwidth save hoti hai aur performance better hoti hai.



```
<picture>
  <!-- Large screens ke liye WebP -->
  <source media='(min-width: 1024px)'
    srcset='hero-large.webp' type='image/webp'>

  <!-- Medium screens ke liye -->
  <source media='(min-width: 768px)'
    srcset='hero-medium.webp' type='image/webp'>

  <!-- Fallback — old browsers ke liye -->
  <img src='hero-small.jpg' alt='Hero Image'
    width='400' height='300'>
</picture>

<!-- srcset — ek hi image, alag resolutions -->
<img src='photo-400.jpg'
  srcset='photo-400.jpg 400w,
    photo-800.jpg 800w,
    photo-1200.jpg 1200w'
  sizes='(max-width: 600px) 400px,
    (max-width: 900px) 800px,
    1200px'
  alt='Responsive photo'>
```

<picture>	Multiple sources container — browser best one choose karta hai
<source>	Ek specific media source — conditions ke saath
media	Screen size condition — min/max-width
srcset	Alag resolutions/sizes ke liye image URLs
sizes	Kaunsi screen size pe kaunsa image size use ho
type	Image format — image/webp, image/avif etc.

■■ Picture tag mein hamesha last mein likho as fallback — yeh zaroori hai, bina iske picture tag kaam nahi karega!



Semantic Tags in HTML

Meaningful tags — page ka structure clearly define karo

KYA HAI? Semantic tags woh elements hain jinke naam se hi pata chalta hai unka purpose kya hai. `<div>` aur `<span>` non-semantic hain — unse kuch pata nahi chalta. Semantic tags se browsers, search engines aur screen readers sab samajhte hain.

■ Semantic vs Non-Semantic

```
<!-- NON-SEMANTIC — koi meaning nahi -->
<div id='header'>...</div>
<div id='nav'>...</div>
<div id='main'>...</div>
<div id='footer'>...</div>

<!-- SEMANTIC — meaning clear hai -->
<header>...</header>
<nav>...</nav>
<main>...</main>
<footer>...</footer>
```

■ Complete Semantic Tags List

<code><header></code>	Page ya section ka top area — logo, heading, nav
<code><nav></code>	Navigation links ka group — menu, breadcrumbs
<code><main></code>	Page ka primary content — sirf ek baar per page
<code><section></code>	Thematically related content ka group — heading ke saath
<code><article></code>	Self-contained, shareable content — blog post, news
<code><aside></code>	Main content se related lekin alag — sidebar, ads, tips
<code><footer></code>	Page ya section ka bottom — copyright, links, contact
<code><figure></code>	Media element wrapper — image, video, diagram, code
<code><figcaption></code>	Figure ka caption — figure ke andar hota hai
<code><details></code>	Collapsible content — click karo toh expand ho
<code><summary></code>	Details tag ka visible heading — clickable title
<code><mark></code>	Highlighted/important text — search results mein useful
<code><time></code>	Date ya time — machine-readable format mein

<code><address></code>	Contact information — author ya organization ka
<code><cite></code>	Kisi creative work ka title — book, film, song
<code><abbr></code>	Abbreviation — title attribute mein full form
<code><progress></code>	Task completion progress — loading bar jaisa
<code><meter></code>	Scalar measurement — disk usage, ratings

■ Details & Summary — Collapsible Content

```
<details>
  <summary>FAQ: HTML kya hai? (click karo)</summary>
  <p>HTML ek markup language hai jo web pages banane ke liye
    use hoti hai. Iska full form HyperText Markup Language hai.</p>
</details>

<details open> <!-- 'open' attribute se default open rahega -->
  <summary>Important Note</summary>
  <p>Yeh content pehle se visible hoga!</p>
</details>
```

■ Time, Mark, Progress & Meter

```
<!-- Time tag — dates machine-readable banao -->
<p>Event: <time datetime='2024-12-25'>25 December 2024</time></p>
<p>Meeting: <time datetime='14:30'>2:30 PM</time></p>

<!-- Mark tag — text highlight karo -->
<p>Search result mein <mark>HTML</mark> word highlighted dikhega.</p>

<!-- Progress tag — loading/completion bar -->
<label>Course progress:</label>
<progress value='75' max='100'>75%</progress>

<!-- Meter tag — measurement/rating -->
<label>Disk usage:</label>
<meter value='6' min='0' max='10' low='3' high='8' optimum='5'>
  6 out of 10 GB
</meter>

<!-- Address tag -->
<address>
  Written by <a href='mailto:harry@example.com'>CodeWithHarry</a><br>
  Visit us at: YouTube.com/CodeWithHarry
</address>
```

■ **Tip:** Semantic tags use karne se SEO improve hota hai, accessibility better hoti hai aur code zyada readable hota hai. Jahan bhi possible ho, `<div>` ki jagah semantic tag use karo!



HTML Entities & Code Tags

Special characters aur code display karne ke liye

KYA HAI? HTML Entities special characters ko display karne ka tarika hai. Kuch characters jaise `<`, `>`, `&` ko directly likhne se HTML toot jaata hai — isliye unhe entity form mein likhte hain.

Most Used HTML Entities

Character	Entity Name	Entity Number	Use Case
<	<	<	Less than — HTML tags mein
>	>	>	Greater than — HTML tags mein
&	&	&	Ampersand — URLs, text mein
"	"	"	Double quote — attributes mein
'	'	'	Apostrophe/single quote
(space)	 	 	Non-breaking space — text mein gap
©	©	©	Copyright symbol
®	®	®	Registered trademark
™	™	™	Trademark symbol
€	€	€	Euro currency
₹	₹	₹	Indian Rupee symbol
♥	♥	♥	Heart symbol
★	★	★	Star symbol — ratings mein
→	→	→	Right arrow
←	←	←	Left arrow
—	—	—	Em dash — sentence break

Code Tags — Code Display karne ke liye

KYA HAI? HTML mein code aur technical content display karne ke liye dedicated tags hain. Yeh tags semantic meaning dete hain aur monospace font mein render hote hain.

```
<!-- Inline code — sentence ke beech mein -->
<p>HTML mein <code>&lt;p&gt;</code> tag paragraph ke liye use hota hai.</p>

<!-- Pre — preformatted, whitespace preserve hota hai -->
<pre>
  Name:   Shauray
  Course: HTML
  Day:    10
</pre>

<!-- Pre + Code — code block ke liye standard way -->
<pre><code>
function greet(name) {
  return 'Hello, ' + name + '!';
}
console.log(greet('Harry'));
</code></pre>

<!-- Keyboard input -->
<p>File save karne ke liye <kbd>Ctrl</kbd> + <kbd>S</kbd> dabao.</p>

<!-- Sample output -->
<p>Program output: <samp>Hello, World!</samp></p>

<!-- Variable -->
<p>Formula: <var>E</var> = <var>m</var><var>c</var><sup>2</sup></p>
```

<code>	Inline code snippet — monospace font mein render hota hai
<pre>	Preformatted text — spaces aur newlines preserve hote hain
<kbd>	Keyboard input — Ctrl+C, Enter jaise shortcuts ke liye
<samp>	Program/computer ka sample output dikhane ke liye
<var>	Mathematical ya programming variable — italic mein dikhta hai

■ **Tip:** Code blocks ke liye hamesha `<pre><code>` combination use karo. Syntax highlighting ke liye Prism.js ya Highlight.js library use kar sakte ho!



Canvas in HTML

JavaScript se browser mein graphics, animations aur games banao

KYA HAI? `<canvas>` ek blank drawing area hai jise JavaScript se control karte hain. Iske andar shapes, text, images draw kar sakte ho aur animations bana sakte ho. Games, charts, image editing — sab canvas se possible hai.

■ Basic Canvas Setup

```
<!-- HTML mein canvas element -->
<canvas id='myCanvas' width='500' height='300'
      style='border: 1px solid #0f3460;'>
  Aapka browser canvas support nahi karta.
</canvas>

<!-- JavaScript se canvas use karna -->
<script>
  const canvas = document.getElementById('myCanvas');
  const ctx = canvas.getContext('2d'); // 2D drawing context

  // Rectangle banana
  ctx.fillStyle = '#e94560';
  ctx.fillRect(50, 50, 200, 100); // x, y, width, height

  // Border wala rectangle
  ctx.strokeStyle = '#00b4d8';
  ctx.lineWidth = 3;
  ctx.strokeRect(300, 50, 150, 100);

  // Rectangle clear karna
  ctx.clearRect(60, 60, 50, 30);
</script>
```

■ Drawing Shapes, Text & Lines

```
const ctx = canvas.getContext('2d');

// CIRCLE banana
ctx.beginPath();
ctx.arc(200, 150, 80, 0, Math.PI * 2); // x, y, radius, start, end
ctx.fillStyle = '#9d4edd';
ctx.fill();
ctx.strokeStyle = '#ffd166';
ctx.lineWidth = 4;
ctx.stroke();

// LINE banana
ctx.beginPath();
ctx.moveTo(0, 0); // starting point
ctx.lineTo(500, 300); // ending point
ctx.strokeStyle = '#06d6a0';
ctx.lineWidth = 2;
ctx.stroke();

// TRIANGLE banana
ctx.beginPath();
ctx.moveTo(250, 50);
ctx.lineTo(100, 250);
ctx.lineTo(400, 250);
ctx.closePath();
ctx.fillStyle = '#f77f00';
ctx.fill();

// TEXT likhna
ctx.font = 'Bold 30px Helvetica';
ctx.fillStyle = 'ffffff';
ctx.fillText('Hello Canvas!', 100, 150); // text, x, y

// IMAGE draw karna
const img = new Image();
img.src = 'photo.jpg';
img.onload = () => ctx.drawImage(img, 0, 0, 300, 200);
```

■ Simple Animation — requestAnimationFrame

```

const canvas = document.getElementById('myCanvas');
const ctx = canvas.getContext('2d');
let x = 0; // ball ki position

function animate() {
  ctx.clearRect(0, 0, canvas.width, canvas.height); // screen saaf karo

  // Ball draw karo
  ctx.beginPath();
  ctx.arc(x, 150, 30, 0, Math.PI * 2);
  ctx.fillStyle = '#e94560';
  ctx.fill();

  x += 3; // har frame mein 3px move karo
  if (x > canvas.width) x = 0; // screen ke bahar gaya toh reset

  requestAnimationFrame(animate); // next frame ke liye call karo
}

animate(); // animation shuru karo

```

getContext('2d')	2D drawing context — shapes, text, images
getContext('webgl')	3D graphics ke liye WebGL context
fillRect(x,y,w,h)	Filled rectangle draw karo
strokeRect(x,y,w,h)	Border-only rectangle draw karo
clearRect(x,y,w,h)	Area ko clear/transparent karo
arc(x,y,r,s,e)	Circle ya arc draw karo
beginPath()	Naya path/shape shuru karo
moveTo(x,y)	Pen uthao aur naye point pe lo jaao
lineTo(x,y)	Current point se line draw karo
fillText(text,x,y)	Filled text likhna
strokeText(text,x,y)	Outline-only text likhna
requestAnimationFrame	Smooth animation ke liye — 60fps

■ **Tip:** Canvas responsive banana ho toh CSS mein `width: 100%` set karo aur JavaScript se `canvas.width = canvas.offsetWidth` set karo resize event pe. Charts banane ke liye `Chart.js` library use karo — scratch se zyada easy hai!



Global Attributes in HTML

Yeh attributes kisi bhi HTML element pe use ho sakte hain

KYA HAI? Global attributes woh attributes hain jo HTML ke KISI BHI element pe apply ho sakte hain — chahe <div> ho, <p> ho, ya <button>. Yeh elements ki core functionality aur behavior define karte hain.

id	Element ko unique identity deta hai — CSS/JS targeting ke liye
class	Element ko ek ya multiple groups mein add karta hai
style	Inline CSS styling — style='color:red; font-size:16px'
title	Hover karne par tooltip dikhata hai
hidden	Element hide ho jaata hai — display:none jaisa
tabindex	Tab key se focus order control karo — 0, 1, 2...
accesskey	Keyboard shortcut set karo — accesskey='h'
contenteditable	User directly element ka text edit kar sakta hai
draggable	Element ko drag kiya ja sakta hai — true/false
spellcheck	Spell check on/off karo — true/false
translate	Browser translation allow karo ya nahi — yes/no
dir	Text direction — ltr (left to right) ya rtl (right to left)
lang	Element ki language set karo — en, hi, ur, fr
data-*	Custom data attributes — apna koi bhi data store karo
aria-*	Accessibility attributes — screen readers ke liye
role	Element ka accessibility role define karo

■ Usage Examples

```

<!-- id, class, style, title -->
<p id='intro' class='highlight' style='color:white;'
  title='Yeh intro text hai'>Hover karo mujhpe!</p>

<!-- hidden attribute -->
<div hidden>Yeh user ko nahi dikhega</div>

<!-- contenteditable – user edit kar sakta hai -->
<div contenteditable='true'
  style='border:1px solid #0f3460; padding:10px;'>
  Yeh text directly edit kar sakte ho!
</div>

<!-- draggable -->
<img src='photo.jpg' draggable='true' alt='Drag me!'>

<!-- dir – Urdu/Arabic ke liye RTL -->
<p dir='rtl' lang='ur'>■ ■ ■ ■ ■ ■ ■ ■ ■ ■</p>

<!-- data-* – custom data store karo -->
<button data-user-id='123' data-role='admin'
  onclick='handleClick(this)'>
  Admin Button
</button>

<script>
function handleClick(btn) {
  const userId = btn.dataset.userId; // '123'
  const role = btn.dataset.role;     // 'admin'
  console.log(userId, role);
}
</script>

<!-- tabindex – keyboard navigation order -->
<input tabindex='1' placeholder='Pehle yahan focus'>
<input tabindex='3' placeholder='Teesre yahan'>
<input tabindex='2' placeholder='Doosre yahan'>

```

■ **Tip:** *data-* attributes* bahut useful hain — JavaScript mein element ke saath extra information store kar sakte ho bina hidden inputs ke. Naming: *data-* ke baad kuch bhi likho — *data-user-id*, *data-price*, *data-color*!



Quotation Tags in HTML

Text quotes, citations aur references ke liye dedicated tags

KYA HAI? HTML mein alag-alag tarah ki quotations ke liye specific tags hain. Sirf visual styling ke liye nahi — semantic meaning ke liye bhi zaroori hain.

<blockquote>	Kisi doosre source se liya hua long/block quote — apni line mein
<q>	Inline short quote — browser automatically quotes add karta hai
<cite>	Kisi creative work ka title — book, film, song, website
<abbr>	Abbreviation ya acronym — title mein full form
<address>	Author ya organization ki contact information
<bdo>	Text direction override — ltr/rtl force karo
<bdi>	Text ko surrounding direction se isolate karo

Code Examples

```
<!-- blockquote — long quote, source bhi batao -->
<blockquote cite='https://developer.mozilla.org'>
  <p>HTML is the standard markup language for creating Web pages.</p>
  <footer>- <cite>MDN Web Docs</cite></footer>
</blockquote>

<!-- q — inline short quote -->
<p>Einstein ne kaha tha: <q>Imagination is more important than knowledge.</q></p>

<!-- cite — work ka naam reference karo -->
<p><cite>Harry Potter</cite> J.K. Rowling ne likhi hai.</p>

<!-- abbr — abbreviation with tooltip -->
<p><abbr title='HyperText Markup Language'>HTML</abbr> web ka backbone hai.</p>
<p><abbr title='World Health Organization'>WHO</abbr> ne guidelines jaari ki.</p>

<!-- address — contact info -->
<address>
  Contact: <a href='mailto:harry@codewithharry.com'>harry@codewithharry.com</a><br>
  YouTube: <a href='https://youtube.com/codewithharry'>CodeWithHarry</a>
</address>

<!-- bdo — text direction override -->
<p>Normal text: Left to Right</p>
<p><bdo dir='rtl'>This text goes Right to Left</bdo></p>
```




Obsolete / Deprecated HTML Tags

Purane tags jo ab use nahi karne chahiye — aur unke modern replacements

KYA HAI? Obsolete tags woh HTML elements hain jo HTML5 mein officially deprecated ho gaye hain. Browser inhe abhi bhi support karte hain backward compatibility ke liye, lekin naye projects mein inhe KABHI use nahi karna chahiye.

■ Yeh tags sirf samajhne ke liye hain — naye projects mein inhe use karna strictly avoid karo! CSS inki jagah pe har kaam kar sakti hai aur better hai.

Obsolete Tag	Kya Karta Tha	Modern Replacement
	Font, color, size set karta tha	CSS: font-family, color, font-size
<center>	Content center align karta tha	CSS: text-align: center / margin: auto
 (sirf visual)	Bold text — no semantic meaning	Use for importance
<i> (sirf visual)	Italic text — no semantic meaning	Use for emphasis
<u> (overused)	Underline — confuse karta hai links se	CSS: text-decoration: underline
<strike>	Strikethrough text	<s> ya use karo
<big>	Bada text	CSS: font-size: larger / 1.2em
<small> (layout)	Chota text for layout	CSS: font-size: smaller (semantic use is ok)
<marquee>	Scrolling/moving text	CSS animations / JavaScript
<blink>	Blinking text	CSS: animation with opacity
<frame>	Page ko frames mein divide karna	CSS Grid / Flexbox / <iframe>
<frameset>	Frames ka container	CSS layout use karo
<noframes>	Frames support na ho toh fallback	Completely avoid karo
<acronym>	Acronym text	<abbr> use karo
<applet>	Java applet embed karta tha	JavaScript / Web APIs use karo
<dir>	Directory list	use karo
<basefont>	Page ka default font set karta tha	CSS body { font-family: ... }

■ Correct vs Incorrect Examples

```
<!-- GALAT – Obsolete tags -->
<font color='red' size='5'>Yeh old style hai</font>
<center>Yeh bhi purana hai</center>
<marquee>Yeh text scroll karta tha</marquee>
<b>Sirf bold – no meaning</b>

<!-- SAHI – Modern HTML + CSS -->
<p style='color: red; font-size: 1.5em;'>Modern style</p>
<p style='text-align: center;'>Modern centering</p>
<strong>Bold with semantic importance</strong>
<em>Italic with semantic emphasis</em>

<!-- Strikethrough – old vs new -->
<strike>Purana price</strike>    <!-- GALAT -->
<s>Purana price</s>              <!-- SAHI – visually struck -->
<del>Deleted text</del>          <!-- SAHI – semantic deletion -->
<ins>Inserted text</ins>         <!-- Naya add kiya gaya text -->
```

■ **Tip:** `<del>` aur `<ins>` especially useful hain document editing mein — jaise Wikipedia pe changes dikhane ke liye. `<del>` = removed content (striketthrough), `<ins>` = added content (underline).



Character Sets in HTML

Browser ko batao text kaise decode karna hai — UTF-8 se Unicode tak

KYA HAI? Character set (charset) ek encoding system hai jo define karta hai ki computer mein text kaise store aur display hoga. Galat charset = garbled text — seedha seedha broken characters dikhenge.

■ Charset Declare karna — ZAROORI

■■ Charset meta tag hamesha **<head>** ka **PEHLA** element hona chahiye — title se bhi pehle! Warna browser galat characters display kar sakta hai.

```
<!DOCTYPE html>
<html lang='en'>
<head>
  <!-- SABSE PEHLE yeh line — koi bhi tag se pehle -->
  <meta charset='UTF-8'>

  <!-- Iske baad baaki tags -->
  <title>Meri Website</title>
  <meta name='viewport' content='width=device-width, initial-scale=1.0'>
</head>
```

■ Common Character Sets

Charset	Full Name	Characters	Use Case
UTF-8	Unicode Transformation Format 8-bit	1,112,064+ characters	<i>STANDARD — sabke liye best choice</i>
UTF-16	Unicode Transformation Format 16-bit	Same as UTF-8	Java, Windows internally use karta hai
UTF-32	Unicode Transformation Format 32-bit	Same characters	Fixed width — memory zyada lagti hai
ISO-8859-1	Latin-1 / Western European	256 characters	Old European sites — avoid karo
ASCII	American Standard Code for Info Interchange	128 characters only	Sirf English — very limited
Windows-1252	Western European Windows	256 characters	Old Windows sites — avoid karo

■ UTF-8 Kyun Best Hai?

```
<!-- UTF-8 se yeh sab ek page pe ho sakta hai: -->

<!-- English -->
<p>Hello World!</p>

<!-- Hindi -->
<p>नमस्ते दुनिया!</p>

<!-- Arabic (RTL) -->
<p dir='rtl'>مرحباً بالعالم</p>

<!-- Japanese -->
<p>こんにちは世界</p>

<!-- Emojis bhi! -->
<p>Web Development 🚀 + HTML 📄 = 🌐</p>

<!-- Math symbols -->
<p> $E = mc^2$  —  $\alpha$ ,  $\beta$ ,  $\gamma$ ,  $\delta$ ,  $\pi$ ,  $\Sigma$ ,  $\infty$ </p>
```

UTF-8	Har cheez cover karta hai — English, Hindi, Arabic, Emojis, Math symbols
Variable width	ASCII chars 1 byte, European 2 bytes, Asian/Emoji 3-4 bytes — efficient!
Backward compat	Pure ASCII text UTF-8 mein bhi same hota hai — koi change nahi
Web standard	99%+ websites UTF-8 use karti hain — Google bhi recommend karta hai
lang attribute	HTML tag mein lang='hi' set karo Hindi ke liye — SEO + accessibility

📌 **Tip:** Code editor (VS Code) mein hamesha 'UTF-8' encoding se file save karo. VS Code mein bottom right corner mein encoding dikhti hai — click karke change kar sakte ho!



ARIA & Accessibility in HTML

Screen readers aur disabled users ke liye web accessible banao

KYA HAI? ARIA (Accessible Rich Internet Applications) attributes browser aur assistive technologies ko batate hain ki elements ka kya purpose hai. Yeh un logon ke liye zaroori hai jo screen readers, keyboard navigation ya other tools use karte hain.

■ ■ Rule #1: Native HTML semantic tags use karo pehle — ARIA sirf tab add karo jab native tags kaam na karein. Bad ARIA is worse than no ARIA!

■ ARIA Roles

```
<!-- role attribute - element ka purpose batao -->
<div role="button" tabindex="0">Click Me</div>
<div role="navigation">...</div>
<div role="main">...</div>
<div role="banner">...</div>
<div role="contentinfo">...</div>
<div role="search">...</div>
<div role="alert">Error: Form submit failed!</div>
<div role="dialog" aria-modal="true">Modal content</div>
<div role="tooltip" id="tip1">Helpful hint here</div>

<!-- Landmark roles (semantic tags se better hai) -->
<!-- <header> = role='banner' -->
<!-- <nav>    = role='navigation' -->
<!-- <main>   = role='main' -->
<!-- <footer> = role='contentinfo' -->
```

■ ARIA Properties & States

```

<!-- aria-label: Element ka readable name -->
<button aria-label="Close dialog">X</button>


<!-- aria-labelledby: Doosre element se label lena -->
<h2 id="form-title">Login Form</h2>
<form aria-labelledby="form-title">...</form>

<!-- aria-describedby: Additional description -->
<input type="password" aria-describedby="pwd-hint">
<p id="pwd-hint">Min 8 chars, 1 number, 1 symbol</p>

<!-- aria-hidden: Screen reader se hide karo -->
<span aria-hidden="true">■</span> <!-- Decorative emoji -->

<!-- aria-expanded: Dropdown/accordion state -->
<button aria-expanded="false" aria-controls="menu">Menu</button>
<ul id="menu" hidden>...</ul>

<!-- aria-required, aria-invalid: Form validation -->
<input aria-required="true" aria-invalid="false">

<!-- aria-live: Dynamic content updates announce karo -->
<div aria-live="polite">Cart: 3 items</div>
<div aria-live="assertive">Error: Connection lost!</div>

<!-- aria-current: Current page/step batao -->
<a href="/home" aria-current="page">Home</a>
<li aria-current="step">Step 2 of 4</li>

```

aria-label	Element ka visible text nahi hai toh label dena
aria-labelledby	Kisi existing element se label reference karna
aria-describedby	Extra description — hint ya error message
aria-hidden	Decorative elements ko screen readers se chhupao
aria-expanded	Dropdown/accordion open hai ya closed
aria-live	Dynamic content changes screen reader ko announce karo
aria-current	Navigation mein current page/step/date batao
aria-required	Form field required hai ya nahi
aria-invalid	Form validation — input valid hai ya nahi
aria-disabled	Element disabled state — better than disabled attr
aria-controls	Kaunsa element ye control karta hai
aria-modal	Dialog/modal ke bahar kuch accessible nahi hai



HTML APIs — Built-in Browser Powers

JavaScript ke saath HTML ke powerful built-in APIs

KYA HAI? HTML5 ne kai powerful APIs introduce kiye jo JavaScript se access hote hain. Inse apps bana sakte ho jo location access karein, data store karein, files handle karein aur bahut kuch — bina kisi external library ke.

■ Web Storage API — localStorage & sessionStorage

```
// localStorage — data permanently store hota hai (jab tak clear na karo)
localStorage.setItem('username', 'Shauray');
localStorage.setItem('theme', 'dark');

// Data retrieve karna
const user = localStorage.getItem('username'); // 'Shauray'

// Data delete karna
localStorage.removeItem('username');
localStorage.clear(); // Sab clear

// JSON objects store karna
const settings = { theme: 'dark', lang: 'hi', fontSize: 16 };
localStorage.setItem('settings', JSON.stringify(settings));
const s = JSON.parse(localStorage.getItem('settings'));

// sessionStorage — tab band hone par data delete ho jaata hai
sessionStorage.setItem('cart', JSON.stringify(['item1', 'item2']));
const cart = JSON.parse(sessionStorage.getItem('cart'));
```

■ Geolocation API — User ki Location

```

// User ki current location lena
if ('geolocation' in navigator) {
  navigator.geolocation.getCurrentPosition(
    function(position) {
      const lat = position.coords.latitude;
      const lng = position.coords.longitude;
      const accuracy = position.coords.accuracy; // meters mein
      console.log('Location:', lat, lng);
    },
    function(error) {
      console.error('Error:', error.message);
    },
    { enableHighAccuracy: true, timeout: 5000 }
  );
} else {
  console.log('Geolocation not supported');
}

// Continuous location tracking
const watchId = navigator.geolocation.watchPosition(callback);
navigator.geolocation.clearWatch(watchId); // Stop tracking

```

■ Drag & Drop API

```

<!-- HTML — draggable elements -->
<div id="drag-item" draggable="true">Drag Me!</div>
<div id="drop-zone">Drop Here</div>

// JavaScript — drag events
const item = document.getElementById('drag-item');
const zone = document.getElementById('drop-zone');

item.addEventListener('dragstart', (e) => {
  e.dataTransfer.setData('text/plain', item.id);
});

zone.addEventListener('dragover', (e) => {
  e.preventDefault(); // Default behavior block karo
  zone.style.background = '#0f3460';
});

zone.addEventListener('drop', (e) => {
  e.preventDefault();
  const id = e.dataTransfer.getData('text/plain');
  zone.appendChild(document.getElementById(id));
});

```

■ History API — Browser Navigation Control

```
// URL change karo bina page reload ke (SPA ke liye)
history.pushState({ page: 'about' }, 'About', '/about');
history.pushState({ page: 'contact' }, 'Contact', '/contact');

// Replace current history entry
history.replaceState({ page: 'home' }, 'Home', '/home');

// Back/Forward buttons
history.back();
history.forward();
history.go(-2); // 2 pages back

// Back button press handle karna
window.addEventListener('popstate', (e) => {
  console.log('State:', e.state);
  // Load content based on state
});
```

localStorage	Permanent storage — browser band karne ke baad bhi rehta hai (5MB)
sessionStorage	Tab-level storage — tab band hone par delete (5MB)
Geolocation	User ki lat/Ing — permission required, HTTPS pe hi kaam karta hai
Drag & Drop	Elements ko drag karke doosri jagah drop karna
History API	URL change karo bina reload ke — Single Page Apps ke liye
Clipboard API	navigator.clipboard.writeText() — copy/paste functionality
Notification API	Browser notifications — permission ke baad desktop pe show
Vibration API	navigator.vibrate(200) — mobile devices ko vibrate karo
Battery API	navigator.getBattery() — device battery status
Network Info	navigator.connection — online/offline, connection speed



Meta Tags — Complete Reference

Head section ke saare meta tags — SEO, Social, Mobile, Security

KYA HAI? <meta> tags <head> mein hote hain aur page ke baare mein information browsers, search engines aur social media platforms ko dete hain. Yeh page pe visible nahi hote lekin bahut important hain.

■ Complete Meta Tags Reference

```
<head> <!-- Part 1: Essential, SEO, OG, Twitter -->
<!-- ■■■ ESSENTIAL ■■■ -->
<meta charset='UTF-8'>
<meta name='viewport' content='width=device-width, initial-scale=1.0'>
<title>Page Title – 50-60 chars ideal</title>

<!-- ■■■ SEO ■■■ -->
<meta name='description' content='Page description 150-160 chars'>
<meta name='keywords' content='html, css, web dev'>
<meta name='author' content='Shauray Sharma'>
<meta name='robots' content='index, follow'>
<meta name='robots' content='noindex, nofollow'> <!-- Indexing band -->
<link rel='canonical' href='https://yoursite.com/page'>

<!-- ■■■ OPEN GRAPH (Facebook, WhatsApp, LinkedIn) ■■■ -->
<meta property='og:title' content='Page Title'>
<meta property='og:description' content='Description'>
<meta property='og:image' content='https://site.com/img.jpg'>
<meta property='og:image:width' content='1200'>
<meta property='og:image:height' content='630'>
<meta property='og:url' content='https://yoursite.com'>
<meta property='og:type' content='website'>
<meta property='og:site_name' content='My Website'>
<meta property='og:locale' content='en_US'>

<!-- ■■■ TWITTER CARD ■■■ -->
<meta name='twitter:card' content='summary_large_image'>
<meta name='twitter:site' content='@yourhandle'>
<meta name='twitter:title' content='Page Title'>
<meta name='twitter:description' content='Description'>
<meta name='twitter:image' content='https://site.com/img.jpg'>

</head>
```

■ Meta Tags Reference — Mobile, Security & Performance

```

<head> <!-- Part 2: Mobile, Security, Performance, Icons -->
  <meta name='theme-color' content='#0f3460'>
  <meta name='apple-mobile-web-app-capable' content='yes'>
  <meta name='apple-mobile-web-app-title' content='My App'>
  <link rel='apple-touch-icon' href='icon-180.png'>
  <link rel='manifest' href='/manifest.json'>

  <!-- ■■■ SECURITY ■■■ -->
  <meta http-equiv='X-UA-Compatible' content='IE=edge'>
  <meta http-equiv='Content-Security-Policy'
    content="default-src 'self'">
  <meta name='referrer' content='no-referrer'>

  <!-- ■■■ PERFORMANCE ■■■ -->
  <link rel='preconnect' href='https://fonts.googleapis.com'>
  <link rel='dns-prefetch' href='https://cdn.example.com'>
  <link rel='preload' as='font' href='font.woff2' crossorigin>
  <link rel='preload' as='image' href='hero.webp'>

  <!-- ■■■ ICONS ■■■ -->
  <link rel='icon' type='image/svg+xml' href='favicon.svg'>
  <link rel='icon' type='image/png' href='favicon.png'>
  <link rel='shortcut icon' href='favicon.ico'>
</head>

```

charset	UTF-8 — text encoding, ALWAYS first in head
viewport	Mobile responsive — width=device-width, initial-scale=1.0
description	Google SERP mein dikhne wala summary — 150-160 chars
robots	index/noindex, follow/nofollow — Google ko instructions
canonical	Duplicate pages ka original URL batao
og:image size	1200x630px ideal — Facebook, WhatsApp share preview
twitter:card	summary / summary_large_image / app / player
theme-color	Mobile browser ka address bar color change karo
manifest	PWA ke liye — app install karne ki capability
Content-Security-Policy	XSS attacks se protection — allowed sources define karo
dns-prefetch	DNS lookup pehle se karo — page load faster
preload	Critical resources priority pe load karo



HTML Form Validation — Advanced

Browser-native validation — bina JavaScript ke forms validate karo

KYA HAI? HTML5 built-in form validation provide karta hai — JavaScript likhne ki zaroorat nahi basic validation ke liye. Browser khud error messages dikhata hai aur form submit rok deta hai.

■ Validation Attributes Complete List

```
<!-- required – field empty nahi hona chahiye -->
<input type='text' name='name' required>

<!-- minlength / maxlength – character count limits -->
<input type='text' minlength='3' maxlength='50'>
<textarea minlength='10' maxlength='500'></textarea>

<!-- min / max – number/date range -->
<input type='number' min='1' max='100' step='5'>
<input type='date' min='2024-01-01' max='2024-12-31'>
<input type='range' min='0' max='10' step='0.5'>

<!-- pattern – regex validation -->
<!-- Indian phone number -->
<input type='tel' pattern='[6-9][0-9]{9}'
      title='10 digit Indian mobile number'>

<!-- Password strength -->
<input type='password'
      pattern='(?=.*[A-Z])(?=.*[0-9])(?=.*[!@#]).{8,}'
      title='Min 8 chars, 1 uppercase, 1 number, 1 symbol'>

<!-- Only letters -->
<input type='text' pattern='[A-Za-z]+' title='Letters only'>

<!-- Indian PIN code -->
<input type='text' pattern='[1-9][0-9]{5}' title='6-digit PIN'>

<!-- novalidate – form pe HTML validation band karo -->
<form novalidate>  <!-- JS se karna ho toh -->
  <input type='email' required>
</form>
```

■ CSS — Validation States Style karna

```
/* Valid input – green border */
input:valid {
  border-color: #06d6a0;
  outline-color: #06d6a0;
}

/* Invalid input – red border */
input:invalid {
  border-color: #e94560;
}

/* Required field jo abhi empty hai */
input:required:placeholder-shown {
  border-color: #ffd166; /* Yellow – fill karo */
}

/* User ne type kiya aur phir bhi invalid hai */
input:not(:placeholder-shown):invalid {
  border-color: #e94560;
  background: #1a0a0a;
}

/* Focus state */
input:focus:valid { box-shadow: 0 0 0 3px rgba(6,214,160,0.3); }
input:focus:invalid { box-shadow: 0 0 0 3px rgba(233,69,96,0.3); }
```

■ Custom Error Messages — JavaScript se

```

<form id='myForm'>
  <input type='email' id='email' required
    placeholder='Email address'>
  <span id='email-error' class='error'></span>
  <button type='submit'>Submit</button>
</form>

<script>
const emailInput = document.getElementById('email');
const errorSpan = document.getElementById('email-error');

emailInput.addEventListener('input', () => {
  if (emailInput.validity.valueMissing) {
    errorSpan.textContent = 'Email dalna zaroori hai!';
  } else if (emailInput.validity.typeMismatch) {
    errorSpan.textContent = 'Valid email address daalo!';
  } else {
    errorSpan.textContent = ''; // No error
  }
});

// Custom message set karna
emailInput.setCustomValidity('Yeh email already registered hai!');
emailInput.setCustomValidity(''); // Clear custom error
</script>

```

validity.valueMissing	required field empty hai
validity.typeMismatch	Email/URL format galat hai
validity.patternMismatch	Pattern se match nahi karta
validity.tooShort	minlength se kam characters
validity.tooLong	maxlength se zyada characters
validity.rangeUnderflow	min se kam value
validity.rangeOverflow	max se zyada value
validity.valid	Sab theek hai — true/false



Responsive HTML — Mobile-First Design

Ek hi HTML code — mobile, tablet, desktop sab pe perfect

KYA HAI? Responsive web design matlab ek hi webpage alag screen sizes pe automatically adjust ho jaaye. Mobile-first approach mein pehle mobile ke liye design karte hain phir bade screens ke liye.

■ Viewport Meta Tag — Most Important

```
<!-- Bina iske mobile pe zoom out ho jaata hai -->
<meta name='viewport' content='width=device-width, initial-scale=1.0'>

<!-- Explained: -->
<!-- width=device-width      : Page width = device screen width -->
<!-- initial-scale=1.0      : Zoom level = 1 (no zoom) -->

<!-- Advanced options (use with caution) -->
<!-- minimum-scale=1.0      : Min zoom allowed -->
<!-- maximum-scale=5.0      : Max zoom allowed -->
<!-- user-scalable=no       : User zoom disable (AVOID — accessibility) -->
```

■ Responsive Images

```
<!-- CSS se image responsive banana -->
<style>
  img { max-width: 100%; height: auto; }
</style>

<!-- srcset – browser best image choose karta hai -->
<img
  src='photo-800.jpg'
  srcset='photo-400.jpg 400w,
          photo-800.jpg 800w,
          photo-1600.jpg 1600w'
  sizes='(max-width: 480px) 400px,
         (max-width: 960px) 800px,
         1600px'
  alt='Responsive photo'>

<!-- Picture tag – art direction -->
<picture>
  <source media='(max-width: 600px)' srcset='mobile-crop.jpg'>
  <source media='(max-width: 1024px)' srcset='tablet-crop.jpg'>
  <img src='desktop-full.jpg' alt='Team photo'>
</picture>

<!-- Responsive video embed -->
<div style='position:relative; padding-bottom:56.25%;'>
  <iframe style='position:absolute; width:100%; height:100%;'
    src='https://youtube.com/embed/VIDEO_ID'></iframe>
</div>
```

■ HTML Structure for Responsive Layout



```
<!DOCTYPE html>
<html lang='en'>
<head>
  <meta charset='UTF-8'>
  <meta name='viewport' content='width=device-width, initial-scale=1.0'>
  <title>Responsive Page</title>
</head>
<body>
  <header>
    <nav>
      <!-- Hamburger menu for mobile -->
      <button class='menu-toggle' aria-label='Toggle menu'>☰</button>
      <ul class='nav-links'>
        <li><a href='/'>Home</a></li>
        <li><a href='/about'>About</a></li>
      </ul>
    </nav>
  </header>

  <main>
    <!-- Fluid container -->
    <div class='container'>
      <section class='hero'>
        <h1>Main Heading</h1>
        <!-- Responsive image -->
        <img src='hero.jpg' alt='Hero' style='max-width:100%'>
      </section>

      <!-- Card grid – CSS Grid/Flex se responsive hoga -->
      <section class='cards'>
        <article class='card'>Card 1</article>
        <article class='card'>Card 2</article>
        <article class='card'>Card 3</article>
      </section>
    </div>
  </main>

  <footer>
    <p>© 2024 My Website</p>
  </footer>
</body>
</html>
```

■ **Tip:** Mobile-first approach: CSS mein pehle mobile styles likho (no media query), phir @media (min-width: 768px) se tablet/desktop styles add karo. Yeh performance ke liye bhi better hai!



HTML Best Practices & Performance

Professional developer ki tarah HTML likhne ke golden rules

KYA HAI? Yeh woh practices hain jo ek beginner ko professional developer se alag karti hain. Clean, fast, accessible aur maintainable HTML likhne ke liye yeh sab follow karo.

■ Document Structure Best Practices

```
<!-- ■ SAHI structure -->
<!DOCTYPE html>
<html lang='en'> <!-- lang hamesha likho -->
<head>
  <meta charset='UTF-8'> <!-- Pehle yahi -->
  <meta name='viewport' content='width=device-width, initial-scale=1.0'>
  <title>Meaningful Page Title | Site Name</title>
  <meta name='description' content='150 chars description'>
  <!-- CSS link head mein – render blocking nahi hoga -->
  <link rel='stylesheet' href='style.css'>
</head>
<body>
  <!-- Semantic structure -->
  <header>
    <nav aria-label='Main navigation'>
      <a href='/' aria-current='page'>Home</a>
    </nav>
  </header>
  <main>
    <h1>One H1 per page only!</h1>
  </main>
  <footer>...</footer>

  <!-- JS scripts body ke end mein – page fast load -->
  <script src='app.js' defer></script>
</body>
</html>
```

■ Performance Best Practices

```
<!-- 1. Images optimize karo -->
<img src='photo.webp'
      alt='Descriptive alt text'
      width='800' height='600'
      loading='lazy'
      decoding='async'>
      <!-- WebP format use karo -->
      <!-- Alt hamesha -->
      <!-- Size specify karo (CLS fix) -->
      <!-- Lazy load -->
      <!-- Async decode -->

<!-- 2. Critical resources preload karo -->
<link rel='preload' as='font'
      href='font.woff2' crossorigin>
<link rel='preload' as='image' href='hero.webp'>

<!-- 3. Third-party domains preconnect karo -->
<link rel='preconnect' href='https://fonts.googleapis.com'>
<link rel='dns-prefetch' href='https://cdn.example.com'>

<!-- 4. Scripts defer/async karo -->
<script src='analytics.js' async></script> <!-- Non-critical -->
<script src='app.js' defer></script> <!-- Main app -->

<!-- 5. Inline critical CSS (above the fold) -->
<style>
  /* Sirf above-fold ka CSS yahan – rest external file mein */
  body { margin: 0; font-family: sans-serif; }
  header { background: #0f3460; }
</style>
```

■ Common Mistakes — Avoid Karo

■ Galat	■ Sahi	Kyun
Multiple <h1> tags	Sirf ek <h1> per page	SEO aur accessibility dono suffer karte hain
<div> for everything	Semantic tags use karo	Meaning lost — SEO aur screen readers suffer
Images bina alt	Hamesha alt likho	Accessibility fail, SEO miss
Inline styles everywhere	CSS file use karo	Maintenance nightmare, no reuse
JS in <head> no defer	defer/async use karo	Page render block hota hai
No viewport meta	Hamesha add karo	Mobile pe broken layout
Empty alt=""	Decorative image? alt="" okay hai	Empty alt tab sirf decorative ke liye theek hai
Tables for layout	CSS Grid/Flexbox	Tables sirf data ke liye hain
Deprecated tags (etc)	Modern HTML + CSS	HTML5 standards follow karo
No lang attribute	<html lang='en'>	Screen readers aur SEO ke liye zaroori



Advanced Tables in HTML

colspan, rowspan, scope, caption — tables ka poora power

KYA HAI? Basic tables toh seekh liye, ab advanced features dekhte hain — cell merging, accessibility, captions aur complex data presentation.

■ colspan & rowspan — Cell Merging

```
<table>
  <caption>Student Results 2024</caption>
  <thead>
    <tr>
      <th scope='col'>Name</th>
      <th scope='col' colspan='2'>Scores</th>  <!-- 2 columns merge -->
      <th scope='col'>Total</th>
    </tr>
    <tr>
      <th scope='col'></th>
      <th scope='col'>Maths</th>
      <th scope='col'>Science</th>
      <th scope='col'></th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td rowspan='2'>Rohan</td>  <!-- 2 rows merge -->
      <td>95</td>
      <td>88</td>
      <td>183</td>
    </tr>
    <tr>
      <!-- rowspan ki wajah se name cell nahi likha -->
      <td>90</td>
      <td>85</td>
      <td>175</td>
    </tr>
  </tbody>
  <tfoot>
    <tr>
      <td colspan='3'>Class Average</td>
      <td>179</td>
    </tr>
  </tfoot>
</table>
```

■ Accessible Tables — scope & headers



```
<!-- scope attribute – screen readers ke liye -->
<table>
  <caption>Monthly Sales Data</caption>
  <thead>
    <tr>
      <th scope='col'>Month</th>
      <th scope='col'>Product A</th>
      <th scope='col'>Product B</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <th scope='row'>January</th> <!-- Row header -->
      <td>■50,000</td>
      <td>■30,000</td>
    </tr>
    <tr>
      <th scope='row'>February</th>
      <td>■65,000</td>
      <td>■40,000</td>
    </tr>
  </tbody>
</table>

<!-- colgroup – columns ko style karo -->
<table>
  <colgroup>
    <col style='background:#0a1628; width:30%'>
    <col span='2' style='background:#0d1b2a'>
  </colgroup>
  <tr><th>Name</th><th>Age</th><th>City</th></tr>
  <tr><td>Rohan</td><td>22</td><td>Delhi</td></tr>
</table>
```

<caption>	Table ka title/description — accessibility ke liye zaroori
scope='col'	Yeh header apne column ke liye hai
scope='row'	Yeh header apni row ke liye hai
scope='colgroup'	Multiple columns ka header
scope='rowgroup'	Multiple rows ka header (thead/tbody/tfoot)
headers	Complex tables mein — cell ka header ID reference
<colgroup>	Columns ka group — styling apply karna
<col>	Single column — span attribute se multiple columns



Advanced Links & Navigation

Anchor tags ke saare use cases, rel attributes aur best practices

KYA HAI? Links HTML ka core feature hai — ek page se doosre page tak jaana. Lekin links sirf URLs ke liye nahi hain — files download, emails, phone calls, page sections sab kuch possible hai.

■ All Link Types

```
<!-- External link – naye tab mein -->
<a href='https://google.com' target='_blank'
  rel='noopener noreferrer'>Google</a>

<!-- Same page section jump -->
<a href='#contact'>Contact section jaao</a>
<section id='contact'>Contact Form</section>

<!-- Page top pe jaao -->
<a href='#'>Top pe jaao</a>
<a href='#top'>Top pe jaao</a>

<!-- File download -->
<a href='resume.pdf' download>Resume Download karo</a>
<a href='photo.jpg' download='shauray-photo.jpg'>Photo Download</a>

<!-- Email link -->
<a href='mailto:hello@example.com'>Email Bhejo</a>
<a href='mailto:hello@example.com
  ?subject=Hello&body=Kaise ho?'>Email with subject</a>

<!-- Phone call (mobile) -->
<a href='tel:+919876543210'>Call Karo</a>

<!-- WhatsApp link -->
<a href='https://wa.me/919876543210?text=Hello!'
  target='_blank'>WhatsApp karo</a>

<!-- SMS -->
<a href='sms:+919876543210?body=Hello'>SMS Bhejo</a>
```

■ rel Attribute — Security & SEO



```
<!-- noopener noreferrer — ZAR00RI for target='_blank' -->
<!-- noopener: New tab apne opener window access na kare (security) -->
<!-- noreferrer: Referrer info na bheje (privacy) -->
<a href='https://external.com' target='_blank'
  rel='noopener noreferrer'>Safe external link</a>

<!-- nofollow — Google ko link follow na karne kaho -->
<a href='https://ads.com' rel='nofollow'>Sponsored</a>

<!-- sponsored — paid/sponsored link mark karo -->
<a href='https://partner.com' rel='sponsored'>Partner Site</a>

<!-- ugc — User Generated Content (comments, forum posts) -->
<a href='https://usersite.com' rel='ugc nofollow'>User Link</a>

<!-- Multiple rel values combine kar sakte hain -->
<a href='https://site.com' target='_blank'
  rel='noopener noreferrer nofollow'>Link</a>
```

download	File download hogi — attribute value = filename
target='_blank'	Naye tab mein khulega
target='_self'	Same tab (default)
target='_parent'	Parent frame mein
target='_top'	Full window mein
rel='noopener'	_blank ke saath hamesha — tab hijacking prevent karo
rel='noreferrer'	Previous page ka URL na bhejo
rel='nofollow'	Google ye link follow na kare — SEO juice na mile
rel='sponsored'	Paid/advertisement link hai
rel='ugc'	User generated content ka link
hreflang	Link kisi doosri language ke page pe — hreflang='hi'
type	Linked resource ka MIME type — type='application/pdf'

■ **Tip:** *target='_blank'* hamesha *rel='noopener noreferrer'* ke saath use karo — *bina iske security vulnerability hoti hai jisme new tab apne opener page ko control kar sakta hai!*



Complete HTML Page Structure

Ek professional HTML page kaisi hoti hai — start to finish

KYA HAI? Ek complete, production-ready HTML page ka template — saari best practices ke saath. Ise apne har naye project ka starting point banao.

■ Complete Production HTML Template — Head

```
<!DOCTYPE html>
<html lang='en' dir='ltr'>
<head>
  <!-- ■■■ ESSENTIAL ■■■ -->
  <meta charset='UTF-8'>
  <meta name='viewport' content='width=device-width,initial-scale=1'>
  <title>Page Title | Site Name</title>
  <!-- ■■■ SEO ■■■ -->
  <meta name='description' content='Page description here'>
  <link rel='canonical' href='https://yoursite.com/page'>
  <!-- ■■■ SOCIAL ■■■ -->
  <meta property='og:title' content='Page Title'>
  <meta property='og:image' content='https://site.com/og.jpg'>
  <meta name='twitter:card' content='summary_large_image'>
  <!-- ■■■ ICONS ■■■ -->
  <link rel='icon' href='favicon.ico'>
  <link rel='apple-touch-icon' href='apple-icon.png'>
  <!-- ■■■ PERFORMANCE ■■■ -->
  <link rel='preconnect' href='https://fonts.googleapis.com'>
  <link rel='preload' as='image' href='hero.webp'>
  <link rel='stylesheet' href='style.css'>
</head>
```

■ Complete Production HTML Template — Body

```
<body>
  <!-- Skip navigation – accessibility -->
  <a href='#main' class='skip-link'>Skip to main content</a>
  <header role='banner'>
    <nav aria-label='Primary navigation'>
      <ul>
        <li><a href='/' aria-current='page'>Home</a></li>
        <li><a href='/about'>About</a></li>
        <li><a href='/contact'>Contact</a></li>
      </ul>
    </nav>
  </header>
  <main id='main' tabindex='-1'>
    <section aria-labelledby='hero-title'>
      <h1 id='hero-title'>Welcome to My Site</h1>
      <p>Hero section content here.</p>
    </section>
    <section aria-labelledby='features-title'>
      <h2 id='features-title'>Features</h2>
      <article class='card'>
        <figure>
          <img src='feature.webp' alt='Feature 1'
              width='400' height='300' loading='lazy'>
        </figure>
        <h3>Feature Title</h3>
        <a href='/feature-1'>Learn more</a>
      </article>
    </section>
  </main>
  <aside aria-label='Related content'>
    <h2>Related Articles</h2>
  </aside>
  <footer role='contentinfo'>
    <nav aria-label='Footer navigation'>
      <a href='/privacy'>Privacy</a>
      <a href='/terms'>Terms</a>
    </nav>
    <p><small>&copy; 2024 Your Name. All rights reserved.</small></p>
  </footer>
  <script src='app.js' defer></script>
</body>
</html>
```

■ **Tip:** Yeh template save karke rakho — har naye project mein yahi use karo as starting point. Isse tumhara code hamesha professional, accessible aur SEO-friendly rahega!



Special Topics — Scripts, Styles & More

Script, style, link, base, noscript — head ke important tags

■ Script Tag — Complete Guide

```
<!-- External script -->
<script src='app.js'></script>

<!-- defer: HTML parse ke baad execute, order maintain karta hai -->
<script src='app.js' defer></script>

<!-- async: Download parallel, execute hote hi (order nahi) -->
<script src='analytics.js' async></script>

<!-- type='module': ES6 modules support, automatically deferred -->
<script type='module' src='app.mjs'></script>

<!-- Inline script -->
<script>
  console.log('Hello from inline script!');
  document.getElementById('btn').click();
</script>

<!-- JSON data script (no execution) -->
<script type='application/json' id='page-data'>
  {"user": "Shauray", "theme": "dark"}
</script>

<!-- Structured data for SEO (Schema.org) -->
<script type='application/ld+json'>
  {
    "@context": "https://schema.org",
    "@type": "Person",
    "name": "Shauray Sharma",
    "url": "https://yoursite.com"
  }
</script>
```

no attribute	Synchronous — page render rok ke load aur execute karta hai
defer	Async download, DOMContentLoaded ke baad execute — order maintain
async	Async download, download hote hi execute — order nahi
type='module'	ES6 import/export support — automatically deferred

crossorigin	CORS requests ke liye — anonymous / use-credentials
integrity	SRI — CDN file tamper na hui ho check karo
nonce	Content Security Policy ke saath use — inline script allow

■ Link Tag — Stylesheet & More

```
<!-- CSS stylesheet -->
<link rel='stylesheet' href='style.css'>

<!-- Media-specific CSS -->
<link rel='stylesheet' href='print.css' media='print'>
<link rel='stylesheet' href='mobile.css'
      media='(max-width: 600px)'>

<!-- Alternative stylesheet -->
<link rel='alternate stylesheet' href='dark.css' title='Dark Theme'>

<!-- Favicon types -->
<link rel='icon' type='image/svg+xml' href='favicon.svg'>
<link rel='icon' type='image/png' sizes='32x32' href='favicon-32.png'>
<link rel='apple-touch-icon' sizes='180x180' href='apple-icon.png'>

<!-- RSS/Atom feed -->
<link rel='alternate' type='application/rss+xml'
      title='My Blog RSS' href='/feed.xml'>

<!-- Preload critical resources -->
<link rel='preload' as='font' href='font.woff2'
      type='font/woff2' crossorigin>
<link rel='preload' as='script' href='critical.js'>
<link rel='preload' as='style' href='critical.css'>

<!-- Prefetch — next page ke resources -->
<link rel='prefetch' href='/next-page.html'>
<link rel='prefetch' as='image' href='next-hero.webp'>
```

■ Base, NoScript & Other Head Tags

```
<!-- base tag – relative URLs ka base set karo -->
<base href='https://yoursite.com/'>
<!-- Ab saare relative URLs is base se challenge -->
<!-- <a href='about'> = https://yoursite.com/about -->

<!-- base target – saare links ke liye default target -->
<base target='_blank'>

<!-- noscript – JavaScript disabled hone par show karo -->
<noscript>
  <p>Is site ke liye JavaScript enable karo.</p>
  <style>body { background: red; }</style>
</noscript>

<!-- style tag – inline CSS (critical CSS ke liye) -->
<style>
  :root {
    --primary: #0f3460;
    --accent: #e94560;
  }
  body { margin: 0; background: var(--primary); }
</style>

<!-- template tag – reusable HTML templates (JS se clone karo) -->
<template id='card-template'>
  <article class='card'>
    <h3 class='card-title'></h3>
    <p class='card-desc'></p>
  </article>
</template>

<script>
  const tpl = document.getElementById('card-template');
  const clone = tpl.content.cloneNode(true);
  clone.querySelector('.card-title').textContent = 'My Card';
  document.body.appendChild(clone);
</script>
```

<base>	Relative URLs ka base URL aur default target set karo
<noscript>	JavaScript band ho toh yeh show hoga — always include karo
<style>	Inline CSS — critical CSS ke liye, baaki external file mein
<template>	Reusable HTML — screen pe nahi dikhta, JS se clone karo
<slot>	Web Components mein content placeholder

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